

INDUCTION

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Reading: PFPL, §2.4, 2.5, 2.6, 3.1

1 Induction

Recall the rules that generate the natural numbers:

$$\frac{\text{ZERO}}{\text{zero nat}} \qquad \frac{\text{Succ} \quad n \text{ nat}}{\text{succ}(n) \text{ nat}}$$

These rules define all the ways of constructing a natural number. Thus, if we *prove* that the truth of a property is preserved by these two rules, it must be that we have proved it for all natural numbers.

This is called the *principle of induction*.

Stated for the natural numbers, the principle of induction is as follows:

Let $\mathcal{P}$ be a property of the natural numbers. If

- $\mathcal{P}(\text{zero})$, and
- whenever $\mathcal{P}(n)$ we know that $\mathcal{P}(\text{succ}(n))$

then $\mathcal{P}(n)$ for all $n \text{ nat}$.

Every set of rules generates an associated induction principle. Thus, the usual principle of natural number induction is implicitly generated by the rules **ZERO** and **Succ**.

We can use induction to prove results.

Claim 1. If $\text{succ}(n) \text{ nat}$ then $n \text{ nat}$.

To prove this it suffices to prove the following property by induction.

$\mathcal{P}(n)$: “If $n \text{ nat}$ and $n = \text{succ}(x)$ for some x , then $x \text{ nat}$.”

2 Simultaneous induction

The principle of induction also applies to the simultaneous generation of judgements in a natural way. Recall the mutually inductive generation of the judgments $n \text{ even}$ and $n \text{ odd}$ by the rules

$$\frac{\text{EVENZ}}{\text{zero even}} \qquad \frac{\text{ODD} \quad n \text{ even}}{\text{succ}(n) \text{ odd}} \qquad \frac{\text{EVEN} \quad n \text{ odd}}{\text{succ}(n) \text{ even}}$$

The associated induction principle is as follows:

Let $\mathcal{P}$ be a property of even numbers, and let $\mathcal{Q}$ be a property of odd numbers. If

- $\mathcal{P}(\text{zero})$, and
- whenever $n \text{ even}$ and $\mathcal{P}(n)$ we have $\mathcal{Q}(\text{succ}(n))$, and

- whenever n odd and $Q(n)$ we have $P(\text{succ}(n))$,
- then $P(n)$ for all n even, and $Q(n)$ for all n odd.

For example, we can prove that

Claim 2. If n even then either $n = \text{zero}$ or $n = \text{succ}(x)$ where x odd.

We cannot prove this by a simple induction; we need to **strengthen** the inductive hypothesis. The proof amounts to performing simultaneous induction over the following predicates P and Q .

$P(n)$: “If n even then either $n = \text{zero}$ or $n = \text{succ}(x)$ where x odd.”

$Q(n)$: “If n odd then $n = \text{succ}(x)$ where x even.”

The claim itself amounts to $P(n)$ for all n even.